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**Bachelor of Engineering**  
Male  
**Pune, Maharashtra, India**

| Examination     | University | Institute                                                                     | Year    | CGPA/% |
|-----------------|------------|-------------------------------------------------------------------------------|---------|--------|
| Graduation      | SPPU       | Shivnagar Vidya Prasark Mandal College of Engineering, Malegaon(Bk), Baramati | 2024-28 | 9.00   |
| Intermediate/+2 | HSC        | Vitthal Gabaji Kapuskar junior college, Mahalunge Padwal                      | 2023    | 60.17  |
| Matriculation   | SSC        | Chaitanya Vidyalyaya Otur, Junnar                                             | 2021    | 94.40  |

PROFILE SUMMARY

➡️ **Robotics / Mechatronics Intern Aspirant**

Second-year Mechatronics Engineering student focused on control systems, robotics, automation, and simulation-driven engineering. Hands-on experience in MATLAB, Simulink, SolidWorks, ANSYS Workbench, and embedded system development through practical engineering projects involving dynamic system analysis, finite element simulation, and intelligent system design. Interested in autonomous systems, system behavior analysis, and interdisciplinary engineering solutions.

INTERNSHIP EXPERIENCES

- President – AeroBotiX Club | SVPM College of Engineering | Oct 2025 – Present**
  - Leading the college Robotics & Drone Club with 50+ active students.
  - Organized robotics workshops and technical learning initiatives.
  - Coordinated project activities and student mentorship.
- Founder & Technical Lead | AstroMechLab | Remote | December 2025- Present**
  - Founded a technical platform focused on robotics, engineering fundamentals, and space systems.
  - Developed research-driven technical articles and structured learning resources.
  - Promoted engineering concepts through technical communication.
- Robotics Intern | Kodacy | Remote | (July 25-Aug 25)**
  - Virtual Internship — Robotics** (Kodacy & SPACE, 15 Days)
  - Completed a short-term robotics internship covering system fundamentals.
  - Built basic robotics simulations and conceptual models.
  - Explored applications of robotics in space science and exploration.

PROJECTS

- Reaction Wheel Shaft Structural Analysis | SolidWorks, ANSYS Workbench, FEM**
  - Designed a stepped reaction wheel shaft for spacecraft attitude control systems.
  - Performed finite element analysis under combined bending and torsional loading.
  - Achieved maximum von Mises stress of **111.78 MPa** with **Factor of Safety = 2.47**.
  - Validated simulation results analytically using mechanics and beam theory equationsGitHub Link:
- PID Controller Design & System Analysis | MATLAB / Simulink**
  - Designed P, PI, and PID controllers using the Ziegler–Nichols method.
  - Simulated third-order system responses in Simulink.
  - Compared rise time, settling time, and overshoot characteristics.
  - Studied trade-offs between response speed and system stability.GitHub Link: <https://github.com/kunaaal-g/pid-ziegler-nichols-control.git>
- JalSetu Smart Water Level Controller | Arduino**
  - Developed an automated water-level monitoring and control system.
  - Integrated sensors with real-time control logic.GitHub Link: <https://github.com/kunaaal-g/JalSetu-Water-Level-Controller.git>
- Frequency Response & Bode Analysis | MATLAB**
  - Performed complete system stability analysis.
  - Calculated gain margin and phase margin numerically.

- Validated theoretical findings using simulation.

**GitHub Link:** <https://github.com/kunaaal-g/bode-plot-stability-analysis.git>

- **Smart Medicine Reminder System | C**

- Built reminder system with scheduling and file handling.
- Implemented alert and snooze functionality.

**GitHub Link:** <https://github.com/kunaaal-g/Smart-Medicine-Reminder-C.git>

- **Robotic Hand Simulation (Ongoing) -Present**

- Designing robotic hand motion and joint coordination system.
- Exploring kinematics and actuator behavior in simulation environment.

## TECHNICAL SKILLS

- **Programming Languages:** C, Python, MATLAB.
- **Control Systems & Robotics:** PID Control, Root locus & Frequency Domain Analysis, Dynamic system Modelling, Robotics Fundamentals, Sensors & Actuators (Basics), Introductory Kinematics.
- **CAD, Simulation & Engineering Tools:** MATLAB, Simulink, SolidWorks, ANSYS Workbench, AUTOCAD, NI Multisim, Finite Element Analysis (FEM), Arduino
- **Professional Skills:** Technical Documentation, Engineering Analysis, Problem Solving, Team Leadership, Presentation skills.

## COURSES & CERTIFICATIONS

- **Introduction to Robotics on AWS** — AWS Training & Certification
- **Robotics** — Prof. Dilip Kumar Pratihar, IIT Kharagpur (NPTEL)
- **START: Scientific Observation from Space** — ISRO-IIRS
- **MATLAB Onramp**— MathWorks
- **Research and Career Opportunities for Space Science and Technology in India**— ISRO-IIRS

## HONOURS / ACHIEVEMENTS

- **2nd Rank – Science Day 2025**, SVPM College of Engineering: Presented on “**Radio Telescopes**” under the theme Solar System & Space Exploration.
- **Top 10 Performer (Grade A+) – National Space Day Quiz 2025, MacroEdtech™**
- **Star Performer Recognition** — Awarded for outstanding participation and leadership development during “**soft skills training**” by the *Eminence team*. Enhanced confidence, teamwork, and communication skills through case studies and group discussions.

## PUBLICATION

- **Biorobotics in Space: Building Life-Inspired Machines for Extreme Worlds** | NoRCEL Institute | Jan 25, 2026
  - Published article exploring bio-inspired robotic systems for extreme environments.
  - **Publication Link:** The 3E Magazine